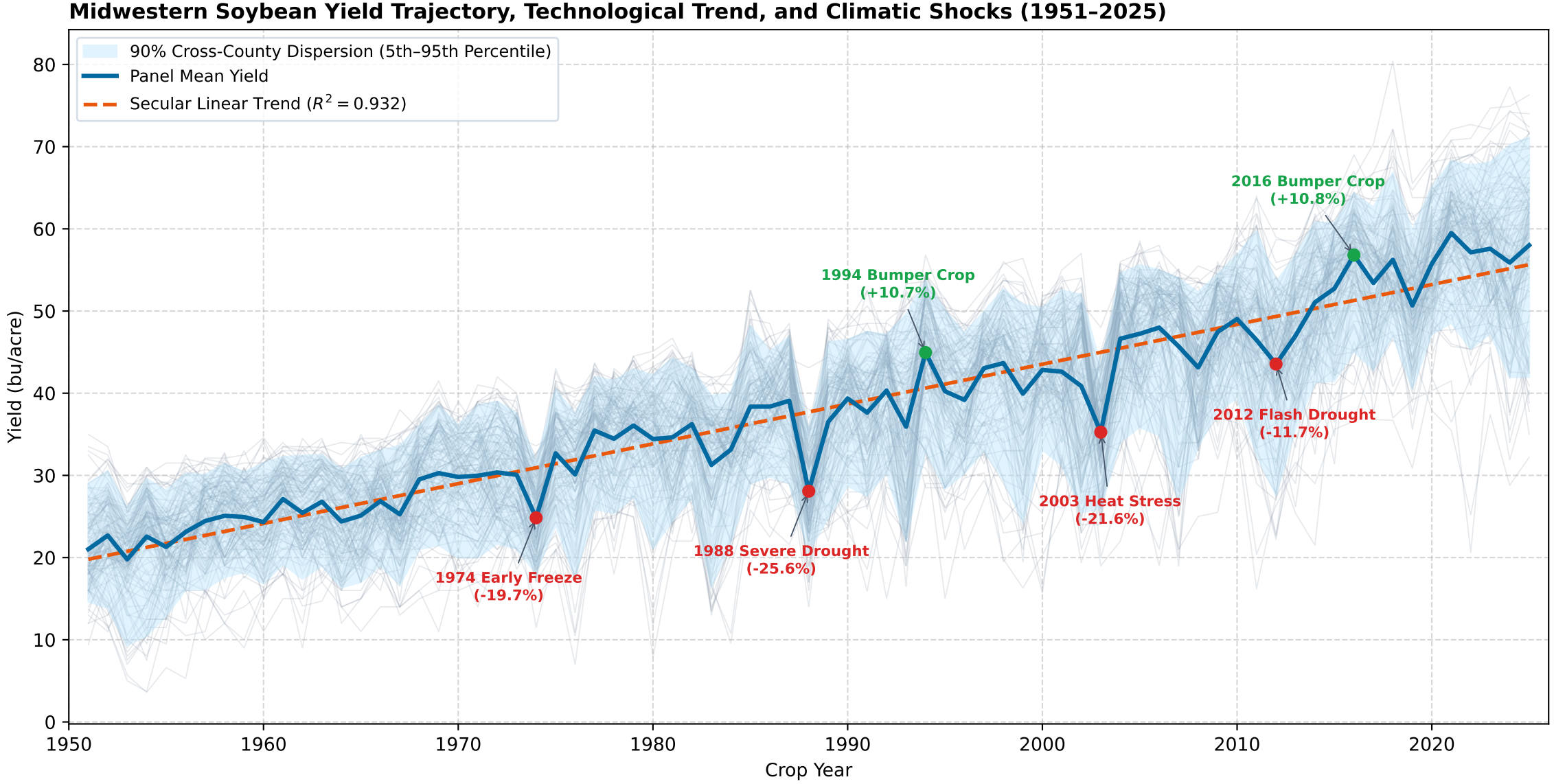

Midwestern Soybean Yield Trajectory, Technological Trend, and Climatic Shocks (1951–2025)
90% Cross-County Dispersion (5th–95th Percentile)
Panel Mean Yield
Secular Linear Trend ($R^2 = 0.932$)
Yield (bu/acre)
Crop Year
0
10
20
30
40
50
60
70
80
1950
1960
1970
1980
1990
2000
2010
2020
1974 Early Freeze
(-19.7%)
1988 Severe Drought
(-25.6%)
1994 Bumper Crop
(+10.7%)
2003 Heat Stress
(-21.6%)
2012 Flash Drought
(-11.7%)
2016 Bumper Crop
(+10.8%)